(10)

$$dI_G = -\frac{c}{2rSl}(1-\rho) I_G S dt = -(1-\rho)\frac{c}{2rl} I_G dt . \quad (13)$$

$$\frac{dI_G}{dt} = -\frac{c(1-\rho)}{2rl} I_G = -\frac{1}{T} I_G , \quad (14)$$

$$T = \frac{2rl}{c(1-\rho)} = 3,00 \cdot 10^{-9} s . \quad (15)$$

3.9 , .

,

$$dN = 2 I_G \sigma_E n \gamma V dt = 2 I_G \sigma_E n \gamma S l dt . \quad (16)$$

$n\gamma V$, $V = Sl$.

(12)

$$\frac{dI_G}{dt} = \frac{\gamma c \sigma_E}{r} n I_G = K n I_G , \quad (17)$$

$$K = \frac{\gamma c \sigma_E}{r} = 5,72 \cdot 10^{10} s^{-1} . \quad (18)$$

정상 발진 상태

3.10

(14) (17) :

$$\frac{dI_G}{dt} = K n I_G - \frac{1}{T} I_G . \quad (19)$$

$$\frac{dn}{dt} = I_0 \sigma_A (1-n) - \frac{1}{\tau} n - 2 I_G \sigma_E n , \quad (20)$$

.

3.11 (19)

.

$$n_{th} = \frac{1}{KT} = 5,83 \cdot 10^{-3} \square 1 . \quad (21)$$

3.12 ,

(20) $I_G = 0$

$$I_{0,th} = \frac{n_{th}}{\tau \sigma_A (1-n_{th})} \approx \frac{n_{th}}{\tau \sigma_A} = 3,58 \cdot 10^{21} cm^{-2} \cdot s^{-1} . \quad (22)$$

(22)

$$\varepsilon = \frac{hc}{\lambda} = 3,83 \cdot 10^{-19} J , \quad (23)$$